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Levels: 300

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Paper 1 of 4 — PEE 314 PETROLEUM GEOLOGY 2024/2025

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF PETROLEUM ENGINEERING

f. PROGRAMME: B.ENG IN PETROLEUM ENGINEERING

jf SESSION: 2024/2025. SEMESTER: FIRST SEMESTER EXAMINATION

f / "COURSE TITLE: [PETROLEUM GEOLOGY | COURSE UNIT: _] 3 oe |

W# (COURSE CODE: | PEE 314 DATE OF EXAM: | 17th MAY, 2025

j

SECTION A (Answer any 2 questions)

Wg. What is a reservoir rock? (5 Marks)

b* Write your notes on the physical properties of a reservoir. (10 Marks)

2. a. Briefly discuss the concept of petroleum migration. (9 Marks)

b. Define porosity and differentiate between effective and total porosity

(10 Marks)

Draw a diagram.

3. Write short notes on the following:

i, Fracture porosity. (3 Marks)

ii Sorting. (3 Marks)

iii. Compaction. (2 Marks)

iv. Cementation. (2 Marks)

v, Water saturation. (5 Marks)

SECTION B (Answer all questions in this section)

4(a). (i) What are the processes followed by oil and gas companies in the production and exploration of

petroleum/crude oil? (5 Marks)

(ii) | What is the origin of petroleum, and what does the term "petroleum" mean? (2 Marks)

(iii) List five logging tools commonly used in the oil and gas industry and the parameters each measure. (10 Marks)

4(b) (i) | Provide some examples (3) of reservoir rock. (3 Marks)

(ii) Briefly explain how crude oil is found and extracted from a reservoir rock. (5 Marks)

SECTION C (Answer only one question)

5(a). Define reservoir trap and state two functions of a reservoir trap. (5 Marks)

(b). State two examples of each of the following traps, providing well-labeled diagrams:

i. Structural trap

ii. Stratigraphic trap (10 Marks)

6(a). Calculate the stock tank oil initially in place and the gas initially in place given the following parameters: Area = 2,000 acres; Thickness = 40 ft; Porosity = 15%; Initial water saturation = 20%; Initial oil formation volume factor = 1.23 bbl/stb; Initial gas formation volume factor = 0.0075 cf/scf. (8 Marks)

(b). A reservoir has a pore volume of 750 cu ft, with an oil saturation of 15%, What is the volume of oil occupying the pore space? (7 Mark)

Paper 2 of 4 — PEE 314 Petroleum Geology

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
2023/2024 First Semester Examinations

PEE 314: PETROLEUM GEOLOGY Time Allowed: 3 hours

SECTION A (Answer any TWO Questions)

- ✓a. What is a Reservoir rock? (5 Marks)
- ✓b. Write short notes on the physical properties of a Reservoir (10 Marks)
- ✓2a. Briefly discuss the concept of Petroleum Migration (5 Marks)
- ✓2b. What is Porosity? (3 Marks)
- ✓2c. Using a diagram, differentiate between effective and total porosity. (7 Marks)
3. Write short notes on the following:
 - i. Fracture porosity (3 Marks)
 - ii. Sorting (3 Marks)
 - iii. Compaction (3 Marks)
 - iv. Cementation (3 Marks)
 - v. Water saturation (3 Marks)

SECTION B (Answer ALL Questions)

- 4ai. Explain the processes that oil and gas companies typically follow for the exploration and production of petroleum and discuss how these processes relate to the field of petroleum geology. (7 Marks)
- 4aii. Explain the origin of petroleum and provide a comprehensive definition of the term "petroleum" in the context of petroleum geology. (3 Marks)
- 4bi. State five logging suites you know (2.5 Marks)
- 4bii. Give three Examples of Reservoir Rock (3 Marks)
- 4c. Briefly explain how crude oil is found and extracted from a reservoir rock (4.5 Marks)

SECTION C (Answer only ONE Question)

- 5ai. Define the Reservoir trap and state two functions of the reservoir trap. (4 Marks)
- 5aii. Describe the key differences between structural and stratigraphic traps in hydrocarbon reservoirs. (5 Marks)
- 5b. State two examples of each of the following traps with a labelled diagram: (i) Structural trap, (ii) Stratigraphic trap? (4 Marks)
- 5ci. Calculate the stock tank oil initially in-place and the gas initially in-place given the following parameters (4 Marks)

Non-effective porosity 39%
29%
effective connected 32%

S. O. 4

Area = 2,000 acres, thickness = 40 ft., porosity = 15%, initial water saturation = 20%, initial oil formation volume factor = 1.23bbl/stb, initial gas formation volume factor = 0.0075cf/scf?

- 5cii. A reservoir having a pore volume of 750cuft. with an oil saturation of 15%. What is the volume of oil occupied in the pore space? (3 Mark)
- 6ai. If the porosity of a reservoir is 25% and the pore volume is 86cuft. Calculate the bulk volume of the reservoir. (3 Marks)
- 6aii. Explain the concept of the petroleum system. Discuss the interplay between source rock, reservoir rock, seal rock, trap, and migration in successful hydrocarbon exploration. (8 Marks)
- 6b. If the API gravity of a particular crude oil is 35°API. Calculate the specific gravity of the crude oil. (2 Marks)
- 6ci. If the specific gravity of a crude oil is 0.685. Calculate the density of the crude oil. (2 Marks)
- 6cii. Explain the primary driving forces behind hydrocarbon migration from source rock to reservoir. (5 Marks)

deep reservoir: They are created by severe faulting in this case hydrocarbon will be rec: Separated with more gas in solution and etc of ~~increased~~ viscosity due to high temp. Porosity and permeability are reduced due to increase compaction.

Paper 3 of 4 — PEE 314 past question

Level 300 • First Semester • PastQuestion

TEST/ASSIGNMENT

PNG 314

INTRODUCTION TO PETROLEUM GEOLOGY

1. Describe the process of Rock Cycle
2. Differentiate between Primary and Secondary migration of hydrocarbon
3. What are the conditions necessary for hydrocarbon accumulation? Explain concisely.
4. What are Faults? List the types of Faults
5. What is the dominant/Mainstream theory of Plate tectonics?
6. Briefly discuss Diagenesis, Catagenesis, and Metagenesis.

Paper 4 of 4 — PEE 314

Level 300 • First Semester • Handout

Overview

Petroleum and natural gas are found within the pore-spaces of porous rock which forms the crude oil and gas reservoirs. Typically, the oil and gas are found with saline water which may be mobile or immobile. This is in contrast to the popular belief that crude oil is found as “oceans of oil” in vast underground caverns. The pore-space in porous rock is identical to the pore-space found within the sand grains of beach sand. If you were to dig a hole in the sand at the beach and fill it with ocean water, then that water would eventually drain out of the bottom of the hole and migrate through the sand back to the ocean. Oil and gas storage and transport are identical to the storage and transport of ocean water in beach sand. In fact, the physical laws that govern the storage and transport of fluids in porous rock and unconsolidated sand are identical.

Learning Objectives

By the end of this lesson, you should be able to:

- illustrate the types of geologic structures where crude oil and natural gas accumulate and/or exit;
- list the five elements that are required to form a hydrocarbon reservoir;
- describe the rock characteristics that are typically associated with hydrocarbons;
- explain the process by which hydrocarbons pass from source rock to reservoir trap;
- describe the geologic properties that make up a reservoir rock and what properties determine the commercial quality of the reservoir rock;
- identify the four hydrocarbon types that make up crude oil and natural gas;
- apply the Gibbs Phase Rule and explain how it governs the phase behavior of complex hydrocarbon mixtures;
- explain how the local hydrostatic gradient and thermal gradient determine the pressure and temperature of a hydrocarbon reservoir at discovery; and
- Summarize how the location of the original pressure and temperature of the reservoir within the phase envelope of the phase diagram determines the type of crude oil or natural gas system that will be produced from the reservoir.

2.1: Basic Earth Geology

Since the occurrence of crude oil and natural gas are intrinsically associated with reservoir rocks, we must start our discussion of the origin and occurrence of natural hydrocarbons with petroleum geology. Geology is the study of all processes that affect the earth. The earth processes studied by geologists occur at many scales from those at the planetary scale (plate tectonics; interactions between the earth’s core, mantle, and crust; etc.) to those at the grain and pore scale (formation of the cementation that binds sand grains, swelling of clays in pores, etc.). Figure 2.01 shows a schematic diagram (not to scale) of the Earth’s Structure.

Through radiometric dating, the earth is believed to be approximately 4.54 billion years old. As shown in Figure 2.01, it is composed of an inner and outer core, the mantle, and the crust. The earth’s crust is the thin, solid, outer shell of the planet that we live on and interact with on a daily

basis. The earth's crust accounts for approximately 1% of the total volume of the planet. The solid crust is the upper layer of the lithosphere (the lithosphere is composed of rigid crust and upper portions of the semi-elastic mantle). The earth's crust can be further divided into the oceanic crust and the continental crust. The oceanic crust is between 3 miles (5 km) and 6 miles (10 km) thick and is overlain by approximately 3 miles of seawater; while the continental crust is between 20 miles (30 km) and 30 miles (50 km) thick. Underlying the oceanic crust and the continental crust is a layer of basalt. Because both the oceanic crust and the continental crust are less dense than the mantle, they both "float" on the mantle.

This floating of the crust on the mantle gives rise to the widely accepted theory of Plate Tectonics. Plate Tectonics is the theory that describes the motion of the continents over the geologic time scale. In this theory, the lithosphere is broken into several major tectonic plates and many smaller plates. The major plates are associated with one of the continents. Slow moving convection currents in the upper mantle are the primary driving mechanisms of plate tectonics; while other mechanisms related to gravity are considered to be secondary driving mechanisms. This is illustrated in Figure 2.02.

In this figure, four tectonic plates are shown along with the convection currents that are driving their drift. The direction of the tectonic drift is shown by the arrows. The separation of tectonic plates is referred to as rifting and is caused by the spreading of two or more convection currents. The mid-Atlantic Ridge is an example of rifting zone. Subduction of tectonic plates, where one plate slides below another plate, results when two or more tectonic plates traveling towards each other collide. The resulting zone is called a subduction zone. A third type of plate boundary, where one plate grinds past another plate, is called a transformational boundary. An example of a transformational boundary is the San Andreas Fault where the Pacific Tectonic Plate is moving across the North American Tectonic Plate.

Due to the friction and built-up stresses in subduction zones and transformational boundaries, these plate boundaries are often associated with volcanism and earthquakes. In fact, the "Ring of Fire" which surrounds the Pacific rim and the North American west coast is caused by the movement of tectonic plates interacting with the Pacific Tectonic Plate.

Rift zones and subduction zones are the only locations where mass transfer from the crust to the mantle (and vice versa) can occur. At a rift zone, fresh rock materials are expelled from the mantle; while at a subduction zone, weathered rock material is returned to the mantle. As we will see, this forms an integral part of the Rock Cycle.

2.2: Rock Types and the Rock Cycle

There are three primary rock types present on the earth: igneous, sedimentary, and metamorphic rocks.

Igneous rocks are rocks formed by the cooling and solidification of molten rock material. As such, igneous rocks are associated with volcanology. There are two types of igneous rocks, intrusive igneous rocks and extrusive igneous rocks. Intrusive igneous rocks crystalize below the earth's surface, allowing for slow cooling and the development of large crystal structures, while extrusive igneous rocks crystalize on the surface, resulting in rapid cooling and the development of small crystal structures. Examples of intrusive igneous rocks include diorite, granite, and peridotite;

while examples of extrusive igneous rocks include basalt, pumice, and tuff.

Sedimentary rocks are rocks formed by the accumulation of sediments. There are three types of sedimentary rocks: clastic sedimentary rocks, chemical sedimentary rocks, and organic sedimentary rocks. Clastic sedimentary rocks are rocks formed by the accumulation of debris that has been mechanically broken by earth processes such as weathering and erosion. Examples of clastic sedimentary rocks include sandstones, siltstones, and shales. Chemical sedimentary rock are rocks that are formed by the precipitation of dissolved materials from high mineral content brines. Examples of chemical sedimentary rocks include rock salt, chert, some limestones, and some dolomites. Finally, organic sedimentary rocks are formed from the accumulation of organic materials (dead plants or animals). Examples of organic sedimentary rocks include coal, some limestones, and some dolomites.

Metamorphic rocks are rocks that started as some other rock type (igneous or sedimentary) but have been substantially altered from their original form by excessive heat, pressure, or exposure to high mineral content brines (or combinations of these three mechanisms). Examples of metamorphic rocks include marble, gneiss, and schist.

As stated in the introduction section of this lesson, crude oil and natural gas reside in the pore-spaces between the grains that make up the rock fabric of the reservoir rock. Consequently, the overwhelming majority of hydrocarbon reservoirs occur in sedimentary rocks where the pore-spaces in the accumulated sediments are sufficiently large to allow for the storage of crude oil and natural gas. While the pore-spaces in metamorphic and igneous rocks are too small to allow for the entry of hydrocarbon molecules, if these rock formations are mechanically fractured by tectonic forces, then the space created by the natural fractures can allow for hydrocarbon storage, and fractured reservoirs in metamorphic and igneous rocks are possible.

It should be emphasized that the formation of rock is not a static process, but is more of a dynamic, cyclical process. In other words: when a rock is formed, it is continually acted upon by earth processes and is subject to change over the geologic time-scale. This process is referred to as the Rock Cycle and is illustrated in Figure 2.03.

2.3: Basic Petroleum Geology

The study of the geological processes that create crude oil and natural gas reservoirs is referred to as Petroleum Geology. In this discussion, we will also need to include brief discussions of related, specialized areas of geologic study including Stratigraphy and Structural Geology. Stratigraphy is the study of the layers (or strata) within rock formations; while Structural Geology is the study of the deformation of rock under tectonic forces. You are probably most familiar with the concepts of stratigraphy and structural geology from road-cuts seen along highways. These road-cuts often show cross-sectional outcrops of layered, deformed rock formations along the road-side. The layering you see is the local geologic stratigraphy; while the deformations you see form the local geologic structure of the rock formations.

Figure 2.04 shows a road-cut along Route 322 approximately 39 miles south of State College, PA. In this photograph, the layering (stratigraphy) and deformation (structure) of the Keefer Formation (Silurian Age) are clearly evident. The “inverted bowl-shaped” feature in this photo is

an anticlinal structure, or simply an anticline. Anticlines statistically form the most common crude oil and natural gas reservoirs. We will discuss this in more detail when we discuss Reservoir Traps later in this lesson.

As we have already discussed, hydrocarbon reservoirs are typically associated with sedimentary rock formations. Over geologic time, weathered and eroded rock materials are carried downstream from elevated regions to lower regions in rivers and streams to oceans, seas, or lakes. At the point in the rivers, oceans, seas, or lakes where the energy in the water can no longer suspend or transport the rock material, it begins to settle in the water and is deposited onto the bottom of the water body. This is called the depositional process. During the depositional process, the environment can change over the geologic time scale: mountain tops and hilltops erode, sea-levels rise or lower, etc. With these changes, the types of sediments that are deposited change and the locations where they are deposited also change. Two commonalities that exist during the depositional process, even over the geologic time scale, are (1) the sediments and, consequently, the sedimentary rocks that are lower in the stratigraphic column were deposited earlier than sediments higher in the stratigraphic column and (2) the layers of sediment are initially deposited horizontally. From this discussion, we can see that sediments lower in the stratigraphic column must be older than those higher in the stratigraphic column.

Note, a stratigraphic column is a written, pictorial, or schematic record of the local rock strata showing the order of the rock deposition (older rocks at the bottom and younger rocks at the top), along with notes concerning the age of the rocks, occurrence of any fossils, any rocks which may be absent from the local geological record due to erosion, or any notes of use to a geologist. One example of a stratigraphic column that you may be familiar with is The Obelisk on the University Park Campus of Penn State. The Obelisk is a physical column constructed of quarried building materials from Pennsylvania with the older rocks placed at the base of the column and the younger rocks placed at the top of the column. Consequently, it forms a true stratigraphic column of the building stones of Pennsylvania. A photograph of The Obelisk is shown in Figure 2.05.

From this discussion, we can also see that in their original, natural state, sedimentary rocks are horizontal. Any deformation (folding, uplifting, tilting, faulting, etc.) in sedimentary rock formations is caused by tectonic forces acting on the rock after deposition and solidification.

There are five geological requirements for the formation of a conventional hydrocarbon reservoir:

1. Source Rock
2. Migration Path
3. Cap Rock
4. Reservoir Rock
5. Trap

These geological requirements are illustrated in Figure 2.06. It needs to be emphasized that although this figure consists of solid colored bands, these bands represent either porous or non-porous rocks. In the following discussions, all of the action concerning this figure is occurring in the pore-spaces of porous rock.

In this figure, the Source Rock is represented by the light grey layer. This is the rock in which the original organic material is converted into hydrocarbons. We will discuss this process later in this lesson. As we can see from this figure, the hydrocarbons do not necessarily originate in the hydrocarbon reservoir itself but are generated away from the reservoir in rocks that are conducive to hydrocarbon generation. As we will discuss later, these source rocks are typically organic-rich shale, siltstones, or coals.

Since the hydrocarbons are generated away from the reservoir, there must be a pathway for the hydrocarbons to migrate from the source to the reservoir. This pathway is the Migration Path in the hydrocarbon system. In the example shown in Figure 2.06, the migration path is a water-filled rock layer (aquifer) that is in communication with both the source rock the hydrocarbon bearing reservoir. This aquifer is shown as the blue layer in Figure 2.06.

In Figure 2.06, the solid green arrows represent the primary migration of the hydrocarbons, while the dashed green arrows represent the secondary migration of the hydrocarbons. Primary migration refers to the initial expulsion of the hydrocarbons from the source rock, while secondary migration refers to the remainder of migration to the reservoir.

As shown in this figure, all fluid migration is upward. This is because the main driving mechanism in hydrocarbon migration is buoyancy, which occurs because the oil and gas are less dense (lighter) than the resident water. In order to prevent this buoyant flow from occurring all of the way to the surface, a vertical flow barrier, or Cap Rock, is required along the migration path and at the reservoir itself. A cap rock is simply an overlying rock layer that is impermeable to flow. Permeability is a property of the rock that is a measure of the ease in which fluids can flow through a porous medium (in our case, a rock formation). In Figure 2.06, the cap rock is depicted by the black layer.

The fourth requirement for a hydrocarbon accumulation is the presence of a Reservoir Rock. In the example shown in Figure 2.06, the reservoir rock is the same rock formation as the migration path. As stated earlier, the most common reservoir rocks are sedimentary rocks; however, naturally fractured igneous and metamorphic rocks can also form hydrocarbon reservoirs. The two requirements for a commercial crude oil or natural gas reservoir are high porosity and high permeability. As discussed earlier, oil and natural gas exist in the pore-space between the grains of the sedimentary rocks. The porosity of a rock is defined as the fraction of the rock's bulk (total) volume occupied by the pores. For example, a cubic foot (1 ft³) of rock with 15 percent porosity will contain 0.15 ft³ of pore space. Consequently, a rock formation with a higher porosity implies greater storage capacity than a rock formation with a lower porosity. This, in turn, results in the possibility of greater quantities of oil and gas stored in the more porous rock.

As discussed earlier, permeability is defined as the ease in which fluids flow through porous media. A high permeability formation implies greater oil and gas production rates and more economically attractive production wells.

The last component of a hydrocarbon system is the Trap. A trap or trapping mechanism is a change in the stratigraphy or a structural deformation that is capable of stopping the migration process and keeping the oil and gas in place over geologic time. In the example in Figure 2.06, the trap is an anticline like that shown in the photograph in Figure 2.04.

Statistically, anticlines are the most common traps; however, they are not the only types of hydrocarbon traps. Crude oil and natural gas traps can be categorized as Stratigraphic Traps or Structural Traps. As the names imply, stratigraphic traps are related to the layering of the rock strata; while structural traps are related to the structural deformations of the rock formations.

Examples of stratigraphic hydrocarbon traps are shown in Figure 2.07. Figure 2.07 shows (a) a “pinch-out” trap and (b) an unconformity trap. In these figures, the yellow layers represent reservoir quality rock (Reservoir Rock); while the green areas represent an oil accumulation in the reservoir trap.

A pinchout trap, Figure 2.07a, is formed due to the relative rise and fall of a water body in relation to the local land mass. (Note: a relative rise or fall may be due to either a true rise or fall of the sea level, or a subsidence or uplift of the land mass.) In this figure, the crude oil reservoir is represented by the green area encased in the black, impermeable layer. As mentioned earlier in this lesson, the size and weight of the sediments that form sedimentary rocks are deposited in water are dependent on the energy of the water suspending the particles. Heavier, coarser materials can only be suspended in high energy environments near the shore where wave and tidal action keep these materials mixed with the water. These materials are deposited as sediments at the point in the depositional environment where the wave and tidal energy can no longer support them. On the other hand, lighter, finer sediments can be suspended in low energy environments away from the shore in more calm waters and, consequently, are deposited further from the shoreline. As the relative position of the sea level changes over the geologic time scale, the positions of the coarser and finer sediments change. The coarser materials with the larger pore spaces between the sediment grains often make good quality rock, while the finer materials, particularly clays and silts, with smaller pore spaces make good impermeable seals (cap rocks). Under the proper sequence of events, as the geographical locations of the sediments change over the geologic time scale, the changes in the size of the sediments can create pinchout, or stratigraphic trap.

Figure 2.07b shows an unconformity trap. A geological unconformity is a boundary between two rock sequences of different ages that are missing some transitory rock sequences of intermediate ages - in other words, some rock strata are missing from the local geological record. This gap in the geological record is caused by a past erosional period.

The depositional history of an unconformity trap is slightly more complicated than that of a pinchout trap. In this trap system, an early depositional period caused sedimentary rocks to be deposited horizontally. These rocks were then tilted due to local tectonic forces. These form the lower tilted layers in Figure 2.07b. The higher, uplifted portions of the tilted layers were then eroded over geological time. The orange wavy line represents an unconformity surface where rocks are missing from the geological record. Geologists recognize these unconformity surfaces when they see rocks of one age sitting on top of rocks of a different age with no intermediary age rocks between them; while paleontologists recognize unconformities by rocks containing fossils from one age sitting directly on top of rocks containing fossils from a different, much older age. After subsidence, a second depositional period occurs where additional sedimentary rocks are

again deposited horizontally over the erosional surface. In order for an unconformity trap to develop, the first layer deposited during the second depositional period must be an impermeable rock which can act as the cap rock for the reservoir. After the second depositional period, the trap can be charged with crude oil to form the oil reservoir - the green area of the figure. (Note that the horizontal yellow layer representing good quality reservoir rock above the unconformity surface, does not contain crude oil or natural gas. This is because it is missing a trapping mechanism at this location.)

These histories illustrate the critical nature of the timing in the development of hydrocarbon reservoirs. For a hydrocarbon reservoir to develop, all five elements of the reservoir system must be in place prior to the formation and migration of the hydrocarbons. If one element is missing from the system, then a crude oil or natural gas reservoir cannot develop.

The second major category of hydrocarbon reservoirs is structural traps. Structural traps, illustrated in Figure 2.08, are related to mechanical deformations and failures of the reservoir rock. Figure 2.08a shows an anticlinal structure, or anticline. As stated earlier, the anticline is statistically the most common reservoir trap. The reason for this is very simple. An anticlinal trap simply requires that the reservoir rock and cap rock be folded in a manner that can trap the lighter hydrocarbons. Figure 2.04 shows the Arch Rock anticline near State College, PA.

Figure 2.08b shows a fault trap. A fault trap is created when a rock formation undergoes mechanical failure during an earthquake and the resulting fault causes a porous and permeable rock formation (reservoir rock) to be positioned adjacent to an impermeable rock formation. Thus, closure to the trap is formed by the cross-fault impermeable layer.

Figure 2.08c shows traps along a salt dome. These traps are common in areas where salt formations exist, such as in the Gulf of Mexico. Since large volumes of salt behave in a plastic manner over geologic time, the weight of the overburden creates great pressures within the salt causing it to flow through any local weak spots in the overburden. These salt intrusions can create hydrocarbon traps above and to the flanks of the intrusion as depicted in Figure 2.08c.

2.4: Origins of Oil and Natural Gas

To this point, we have discussed the elements required for the formation of a hydrocarbon reservoir. We have also discussed the timing and histories of the formation of the reservoir traps. We have also mentioned that all of the elements of the hydrocarbon reservoir must be in place prior to the formation and migration of the hydrocarbons. We are now able to discuss the actual formation of the hydrocarbons themselves.

Petroleum and natural gas are naturally occurring substances composed of hydrocarbon molecules (molecules made up of hydrogen and carbon atoms) and, possibly, non-organic contaminants, such as CO₂, H₂S, N₂, and O₂.

The most widely accepted theory of the origin of petroleum and natural gas is the Organic Origin Theory which states that these fluids are generated from the decay of prehistoric plants and animals under the influence of the excessive pressures and temperatures that exist in the earth's subsurface. Research indicates that petroleum and natural gas originated from zooplankton (single-celled plants found drifting freely in fresh or brackish water) and algae. During their lifetimes, these organisms create energy from photosynthesis to carry out their life processes,

and it is this energy that we use today (thus the term, “fossil fuels”). Contrary to the popular myth, petroleum and natural gas do not originate from decomposed dinosaurs.

Over the geologic time scale, these free-floating zooplankton and algae die, settle to the ocean-, sea-, lake-, river-, or swamp-bottom, and accumulate as sediment. This process occurs simultaneously with the geological processes (erosion, transportation, and deposition) acting on the sand, silt, and clay sediments that form sedimentary rocks. As the organic material and rock materials accumulate on the sea bottom, the materials at the top of the accumulation exert increased weight (pressure) on the materials at the bottom of the accumulation. As this depositional process continues and the accumulation becomes thicker, the impact of the earth’s geothermal gradient also begins to act on the organic materials. At elevated pressures and temperatures, the decaying organic materials are transformed into a dark waxy material called kerogen. Kerogen is an intermediary stage in the development of petroleum. The process of converting the original organic material into kerogen is called the Catagenesis Process. The formation of kerogen requires that the rock containing the original organic material be impermeable. This requirement is essential because as kerogens are formed, they must remain trapped within the pore-spaces of the rock and cannot be allowed to escape to other rock formations or to the environment. As we have already discussed, the rock formation in which the original organic materials are converted to kerogen, and eventually petroleum, are called source rocks for the hydrocarbons. Typically, fine-grained, clay-rich sedimentary rocks, such as shales, are the most common source rocks for hydrocarbon basins because they satisfy two of the requirements for kerogen generation; they (1) are sedimentary rocks and (2) contain adequately low permeabilities.

During the catagenesis process, the kerogen is “cooked” in the high-temperature environment, and the long-chain kerogen molecules are “cracked” into shorter-chained hydrocarbon molecules. It may take several million years for the deposition process to bury the original organic materials and the source rock to achieve the appropriate temperatures ($> 250 - 300^{\circ}\text{F}$) for kerogen generation and another several million years to generate commercial quantities of oil and gas. There is a specific temperature range, referred to as the “oil window,” in which oil is formed. At temperatures below the oil window, the Catagenesis process cannot occur; while at higher temperatures the “cooking” and “cracking” processes are stronger and very short-chain natural gas molecules are generated (thermogenic gas).

A second, less accepted theory for the origin of hydrocarbons is the Inorganic or Abiogenic Theory. In this theory, hydrogen and carbon from inorganic sources are fused at the elevated pressures and temperatures in the earth’s subsurface and are converted to hydrocarbons. While the presence of inorganically sourced hydrocarbons cannot be discounted, there is abundant evidence that the vast majority of crude oil and natural gas have an organic origin.

2.5: Hydrocarbon Types

Crude oil is a complex mixture of several types of hydrocarbon molecules along with inorganic impurities. These hydrocarbon molecules are:

1. Paraffinic (or alkane series) hydrocarbons

2. Naphthenic (or saturated cyclic) hydrocarbons
3. Aromatic (or cyclic) hydrocarbons
4. Asphaltene hydrocarbons

Sometimes we refer to a particular crude oil as a paraffinic crude oil or as an aromatic crude oil. While there may be several thousand different hydrocarbon molecules in a given crude oil, these descriptions simply refer to the dominant hydrocarbon type in the mixture. All naturally occurring crude oils will typically contain molecules of each type.

Paraffinic, or alkane series, hydrocarbons are shown in Table 2.01. These hydrocarbons are typically studied in introductory courses in Organic Chemistry. Alkane series hydrocarbons are composed of only hydrogen and carbon atoms attached with single bonds. They are characterized by the following formula for the number of hydrocarbon atoms present in the molecule:

$$NH = 2NC + 2 \quad \text{Equation 2.01}$$

In this formula, NH is the number of hydrogen atoms, while NC is the number of carbon atoms in the hydrocarbon molecule.

Normal alkanes are the chain molecules depicted the first four examples in Table 2.01. Once the number of carbon atoms reaches four (butanes) different permutations of an alkane molecule can exist that still honor Equation 2.01 but do not form chain molecules. These permutations are called isomers with the number of possible permutations (and number of isomers) increasing as the number of carbon atoms, NC, increases. An example of the difference between a normal and isomeric alkane is shown in Table 2.01 for n-butane and i-butane. Due to the slight discrepancies in the molecular structures, normal alkanes and isomeric alkanes have slightly different physical and chemical properties, such as, boiling points, melting points, etc.

As the number of carbon atoms increases, the molecular weight of the molecule increases. The lower molecular weight alkanes, methane and ethane, are the most common hydrocarbon components in natural gas; however, intermediate molecular weight alkanes up to the butanes may also be present in natural gases. In addition to these hydrocarbon components, some inorganic impurities, such as CO₂, H₂S, N₂, and O₂ may also be present in natural gases.

Crude oils from Pennsylvania are generally classified as Paraffinic Crude Oils ("Pennsylvania Grade Crude Oil" was at one time a phrase used to describe high quality crude oils.). As a solid, the alkanes form a waxy substance, paraffin, which is the main component of such products as paraffin candles.

The second category of hydrocarbon molecules found in crude oils are naphthenic (or saturated cyclic) hydrocarbons. Like the alkanes, these hydrocarbons are composed of hydrogen and carbon atoms attached with single bonds; however, they differ from the alkanes in that they do not form chain structures but form cyclic (ring) structures. Examples of naphthenic hydrocarbons are shown in Table 2.02.

The third category of hydrocarbon molecules found in crude oils are the aromatic (or cyclic) hydrocarbons. These hydrocarbons are composed of hydrogen and carbon atoms that form cyclic structures but contain dual bonds between alternating carbon atoms. The simplest aromatic

hydrocarbon is benzene with a single ring structure; while more complex aromatic hydrocarbons are typically formed with multi-ring structures. As the name implies, aromatic hydrocarbons, such as benzene, are associated with a sweet smell. Examples of aromatic hydrocarbons are shown in Table 2.03.

The last category of hydrocarbon molecules found in crude oils are the asphaltene hydrocarbons. These hydrocarbons are large, high molecular weight molecules which may also contain some atoms other than hydrogen and carbon atoms, such as, sulfur, oxygen, or nitrogen atoms. Asphaltenes typically result in the residue from the refining process and are the principal components of asphalt (road paving), tar, and bitumen products.

The mixture of these hydrocarbon molecules can have several consequences on oil and gas production, transport, and refining. These include:

1. The mixture of hydrocarbon molecules gives the bulk fluids (oil and gas phases) the properties which dictate the ease or difficulty in which they flow through the reservoir towards production wells and through production equipment
2. The nature of any solids which may precipitate from the bulk fluids onto rock grains, wellbore, or surface equipment during production causing operational problems requiring different solutions:
 - a. Paraffinic crude oils may deposit waxy solids onto the reservoir rock, well completion, or production tubing causing restrictions to flow requiring a remediation treatment, such as hot kerosene injection into the well.
 - b. Asphaltic crude oils may deposit hydrocarbon resins onto the reservoir rock, well completion, or production tubing causing restrictions to flow requiring a remediation treatment, such as solvent injection into the well.
3. Local field treatment of the crude oil to upgrade it to the required specifications for the pipelines and refineries where the crude oil will eventually flow
4. The products that can be refined from the crude oils once they reach the refineries: lubricating oils, heating oils, car or aviation fuels, asphalt and tar products, etc.

The definitive methods for determining the different components in crude oil are with laboratory measurements, such as gas chromatography. Often, however, less rigorous methods may be useful for quick, on-site evaluations or for numerical (computer) calculations where the crude oil type is characterized with a single parameter. In the oil and gas industry, one common measure of the dominant character of the crude oil (paraffinic, naphthenic, and aromatic) is the Watson Characterization Factor (or a generalization suggested by Whitson). The original form of the Watson Characterization Factor, KW , is:

$$KW = T_b^{1/3} \gamma_o \quad \text{Equation 2.02a}$$

Where:

KW is Watson Characterization Factor, $^{\circ}R^{1/3}$

T_b is the mean average boiling point of the mixture, $^{\circ}R$

γ_o is the specific gravity of the mixture, dimensionless

This equation requires that the boiling point, T_b , for the mixture of interest is known. In many

cases, T_b is unknown or difficult to measure. For these cases, Whitson correlated the Watson Characterization Factor with the more commonly known molecular weight:

$$K W = 4.5579 M_o^{0.15178} \gamma_o^{-0.84573} \quad \text{Equation 2.02b}$$

Where:

KW is Watson Characterization Factor, $oR^{1/3}$

M_o is the molecular weight of the mixture, $lbm/lbm\text{-}mol$

γ_o is the specific gravity of the mixture, dimensionless

The guidelines for the use of the Watson Characterization Factor are:

1. $KW \geq 12.5$ indicates a paraffinic crude oil
2. $12.5 > KW \geq 10$ indicates a crude oil containing naphthenic or aromatic components
3. $KW < 10$ indicates a highly aromatic crude oil

Table 2.04 shows the values of the Watson Characterization Factor for substances of known hydrocarbon type (paraffinic, naphthenic, and aromatic).

(A) One of the isomers is Hexane

2.6: Types of Hydrocarbon Reservoirs

To this point in the lesson, we have discussed the origins of the reservoir rocks and traps that formed over the geologic time scale which result in modern oil and gas reservoirs. We have also discussed the origins of the hydrocarbons that reside in pore spaces of these reservoirs. We can now discuss the types of reservoirs and reservoir systems that are currently of commercial interest to the oil and gas industry. To aid in this discussion, we must first introduce the concept of the Phase Envelope (or, Phase Diagram, or Pressure-Temperature Diagram, or P-T Diagram) for single-component (pure) and multi-component (mixture) systems.

2.7: The Gibbs Phase Rule

The Gibbs Phase Rule relates the degrees of freedom in a system to the number of components and number of phases in a system. The Gibbs Phase Rule is:

$$F = C - P + 2 \quad F = C - P + 2$$

Equation 2.03

Where:

F the number of degrees of freedom in the system, integer

C is the number of components in the system, integer

P is the number of phases in the system, integer

The use of the Gibbs Phase Rule is best illustrated with examples; however, to do this we must first discuss some fundamental thermodynamic concepts. The phrase “degrees of freedom” refers to the maximum number of independent thermodynamic variables (pressure, temperature, and intensive variables) that you can vary simultaneously within a system at equilibrium before you are forced to specify one or more of the remaining variables (or disturb the equilibrium of the system). By “intensive variables”, we are referring to variables that are independent of the size of the system. For example, phase density in a system is an intensive variable because you can halve the size of the system, and the phase density will

remain the same. On the other hand, the mass of a system is an extensive property because if you halve the size of the system, then you halve the mass in the system. A component refers an individual chemical element that exists in the system (in our case, a molecular species: methane, ethane, cyclopentane, benzene, CO₂, H₂O, etc.). Finally, a phase is a physical state of matter with homogenous (uniform) composition, physical properties, and chemical properties. In petroleum and natural gas engineering, we typically deal with systems containing four phases: a gaseous hydrocarbon phase (natural gas), a liquid hydrocarbon phase (crude oil), a liquid aqueous phase (water or brine), and a solid rock phase. There are times when we deal with systems containing more phases (such a solid hydrocarbon phase or multiple hydrocarbon liquid phases), but the circumstances when this occurs are beyond the scope of this class.

For a single-component system, we have $C = 1$, and Gibbs Phase Rule becomes:

$$F = 3 - P \quad \text{Equation 2.04}$$

Equation 2.04

If we were to plot the phase state (number and types of phases) on a pressure-temperature plot for this single-component system, then we would obtain a plot like that shown in Figure 2.09.

This figure is also referred to as a Phase Diagram or a P-T Diagram. In this Phase Diagram, the grey region represents all of the pressure-temperature combinations that result in the solid form of our single component, the green region represents the pressure and temperature combinations that result in the liquid form of our single component, the red region represents the pressure and temperature combinations that result in the gaseous form of our single component, and the multi-colored region represents the super critical form of our single component. Also posted on this Phase Diagram are two points, the triple point and the critical point.

Getting back to the single component version of the Gibbs Phase Rule, Equation 2.04, we can start to clarify some of the concepts that we have already discussed with some examples. If we consider the single-phase regions in Figure 2.09 ($P = 1$ in Equation 2.04), then from Equation 2.04, we have 2 degrees of freedom ($F = 2$). This implies that we can independently vary Pressure and Temperature within the Phase Envelope (colored regions) and not change the phase state of the system. In other words, two degrees of freedom represent two-dimensional regions (areas) on the Phase Diagram in which a single phase exists.

Now, let's consider the occurrence of two phases coexisting simultaneously ($P = 2$) in equilibrium. From Equation 2.04 with $P = 2$, we have one degree of freedom ($F = 1$). In Figure 2.09, we have two coexisting phases along the borders of the phase envelopes. For example, the line bordering the Solid and Liquid Phases represents the pressure-temperature conditions where the solid form of our single component system can coexist with the liquid form of our single component at equilibrium (think of polar caps sitting on water near the earth's poles). For two phases to coexist in equilibrium, if we were to change one variable, say temperature, then we would be forced to change pressure in a manner that it remained on the border line between the two single-phase regions. In other words, one degree of freedom represents the one-dimensional, curvilinear lines which act as borders between the single-phase regions. If we change one variable on one of these border lines, then we are forced to change the other variable to remain on the border line.

Finally, we can consider the coexistence of three phases ($P = 3$ in Equation 2-4) in a system at equilibrium. From Equation 2.04 with $P = 3$, we have $F = 0$, or no degrees of freedom. This implies that at the point where three phases coexist (the Triple Point in Figure 2.09), we cannot change either pressure or temperature and retain a three-phase system in equilibrium of the single-component (pure) system. The Triple Point is a property of the component that we are considering. Thus, zero degrees of freedom refers to a 0-Dimensional point (the Triple Point). In Figure 2.09, the Critical Point is also plotted. The Critical Point, defined by the critical pressure, P_c , and the critical temperature, T_c : (P_c, T_c) is the point in the system that defines the onset of the super critical state. Super critical fluids are fluids in which the gaseous phase becomes indistinguishable from the liquid phase (the phase densities become equal).

Consider a pressure-temperature pair in the single-phase gas region with a temperature somewhere between the temperature at the triple point and the critical temperature, T_c , such as Point A in Figure 2.10. If we were to increase the pressure of this single-phase gas (Path A-A'), then we would see a distinct and abrupt change in the phase state of the fluid as we crossed into the single-phase liquid region. This is because the densities of the gaseous phase and the liquid phase are different. Now, if we were to do the same experiment starting at a temperature greater the critical temperature, T_c , such as Point B, then as we increased the pressure on the single-phase gas (Path B-B') and entered the Super Critical Fluid region, there would be no abrupt phase change.

We could also perform a third experiment where we started at the original point, Point A, and followed the pressure-temperature path Path A-B-B'-A' into the single-phase liquid region to arrive at Point A'. In this last experiment, we would arrive at the same point, Point A', as in the first experiment, but without any abrupt phase change. The system properties would change smoothly and continuously during the entire experiment.

For multi-component systems like real crude oil – natural gas systems, the Pressure-Temperature Diagram is much more complex. This is because a real crude oil – natural gas system may contain tens or hundreds of thousands of components. For Multi-Component Systems, the P-T Diagram looks like that in Figure 2.11. A Phase Diagram, such as that shown in this figure, typically is measured in the laboratory but can also be generated mathematically with sophisticated models, such as, Cubic Equation of State Equations (multi-component extensions to van der Waal's Equation).

In this figure, the region between the green curve and the red curve is the two-phase envelope; while the region outside of the two-phase envelope is the single-phase region. Single-phase liquids exist to the left (lower temperatures) and above (higher pressures) the bubble-point locus (green curve); while single-phase gases exist to the right (higher temperatures) of the dew-point locus (red curve).

The green curve in this figure represents the Bubble-Point Locus of this multi-component system; the red curve represents the Dew Point Locus of the system, and the dashed lines in the two-phase region represents the quality lines of the system (lines of constant volume-fraction of the liquid phase). The bubble-point, defined by a bubble-point pressure, P_b and a bubble-point temperature, T_b , is the point on a pressure-temperature path (originating in the single-phase

liquid region) where the path enters the two-phase region (crosses the green curve in Figure 2.11). The name “bubble-point” comes from the fact that this is the point where the first bubble of gas evolved from a liquid as it enters the two-phase region. For example, Point A in Figure 2.11 lies in the single-phase liquid region. As the pressure is reduced at a constant temperature (isothermal conditions), it follows the Path A-A'. The point where Path A-A' enters the two-phase region (crosses the green, bubble-point locus) represents the bubble-point pressure for this temperature. This is the point where the first bubble of gas is formed in the system. The pressure reduction is continued until it terminates at Point A'. Point A' lies on the 0.9 quality line, implying that at this point, the system is composed of two-phases with the liquid phase occupying 90 percent of the volume.

On the other hand, the dew-point, defined by the dew-point pressure and dew-point temperature is the point on a pressure-temperature path (originating in the single-phase gas region) enters the two-phase region. The name dew-point refers to the point where the first liquid drop condenses from the gaseous phase. For example, Point B in Figure 2.11 lies in the single-phase gas region and enters the two-phase region when the pressure is reduced under isothermal conditions (Path B-B'). The dew point pressure for this temperature then is the pressure where Path B-B' crosses the red dew-point locus. If the isothermal pressure reduction is stopped at Point B', then the system at equilibrium will contain two phases with the liquid phase occupying 10 percent of the system volume.

Finally, the pressure, $P_{\max}$ in Figure 2.11 is the cricondenbar (the maximum pressure in which two phases can coexist); while the temperature, $T_{\max}$ is the cricondentherm (the maximum temperature in which two phases can coexist).

With these preliminary concepts, we can now continue with our discussion on the types of hydrocarbon reservoirs encountered in the oil and gas industry.

2.8: Undersaturated Black Oil (Low Shrinkage Oil) Reservoirs

Undersaturated black oils, sometimes referred to as “low shrinkage oils,” are single-phase liquid systems that reside in reservoirs with an original temperature that is significantly lower than critical temperature, T_C . The phase diagram for an undersaturated black oil is shown in Figure 2.12.

In this figure, the reservoir temperature, T_R , and reservoir pressure, p_R , at the time of discovery are determined by the earth's local temperature gradient and hydrostatic pressure gradient, respectively. The phase diagram for the crude oil is determined by the composition of the crude oil. As shown in this figure, the original reservoir temperature and pressure lie in the single-phase liquid region of the phase diagram at a temperature lower than the critical temperature.

The term “undersaturated” implies that the crude oil has the capacity to dissolve additional gaseous components and remain in the single-phase region. For example, if we change the composition of the system by adding more lower molecular weight components (e.g., methane and ethane), then the phase envelope would expand with the bubble-point locus beginning to shift towards the Point (T_R , p_R). In this example, we could continue to add gaseous components to the system until the bubble-point locus has shifted all of the way to the Point (T_R , p_R). At this point, with Point (T_R , p_R) lying on the bubble-point locus, we would call the fluid a bubble-point

fluid. This is the point where the system would be unable to accept additional gaseous components and still remain a single-phase liquid, i.e., a single-phase crude oil. We will discuss the significance of the term “low shrinkage” when we discuss “high shrinkage” volatile oils. The production of oil and gas is an isothermal process. We typically deplete the reservoir pressure by removing fluids from the system. Even in the cases where we inject fluids at different temperatures into the reservoir, such as with steam injection, the volume of fluids that we inject are so small compared to the size of the reservoir and the over- and under-burdens that, at most, we cause a local change in the temperature at the point of injection and have no significant impact on the temperature of the system. For isothermal processes, the pressure-temperature path of the fluids inside the reservoir is a vertical line at a fixed temperature. This is represented in Figure 2.12 (A) as the solid black arrow going from Point (TR, pR) to Point (TR, pA).

Figure 2.12 (B) shows the undersaturated black oil in an anticlinal trap at the time of discovery, that is, at the original reservoir temperature, TR, and pressure, pR. This figure depicts a single-phase liquid hydrocarbon (crude oil) system. As we produce reservoir fluids, single-phase oil (and the dissolved natural gas components) and possibly water, the pressure in the reservoir depletes. This is the solid arrow from Point (TR, pR) to Point (TR, pA). At some point in time, the reservoir pressure drops below the bubble-point pressure, and we enter the two-phase region in the phase diagram. At this point in time, a second, free gas phase evolves in the reservoir. This occurs when the path of the reservoir fluids crosses the green, bubble-point locus.

As we continue production, we now produce crude oil (and the gas dissolved in it), free natural gas, and, possibly, water. At some point during the depletion process, the natural pressure in the reservoir becomes so low that it can no longer overcome the weight of the fluids in the wells and lift these fluids to the surface at commercial rates. At that point in time, we may be able to install some form of artificial lift (e.g., pumps) to help the natural reservoir pressure lift the reservoir fluids, but that is only a temporary solution, and eventually we will be forced to abandon the reservoir. The pressure at which we abandon the reservoir is called the “abandonment pressure,” pA.

Figure 2.12 (C) shows the same reservoir at abandonment conditions after the free gas phase has developed. Since gas is lighter than oil, it will migrate to the top of the reservoir and accumulate there. When gas comes out of solution due to pressure depletion and accumulates at the top of a reservoir, we refer to the resulting accumulation as a “secondary gas cap.”

Please note that we may run into some situations in which the abandonment pressure occurs before we drop below the bubble-point pressure. In these cases, the path of the reservoir fluids would remain in the single-phase liquid region, and a free gas phase would never develop in the reservoir.

Also depicted in Figure 2.12 (A) is the dashed arrow from Point (TR, pR) to Point (TS, pS). This is the path that the reservoir fluids travel in the well and through the surface facilities. Point (TS, pS) refers to the separator conditions. The separators are field equipment, typically designed by facilities engineers that are used to separate the produced fluids. Normally, produced oil has a higher commercial value to the oil and gas company than produced natural gas, so typically the facilities engineers design the separators to optimize the volume of liquid at the surface. In other

words, the faculties engineers will design the separator conditions T_S and p_S to lie on the quality line of the phase diagram with the maximum feasible volume percent liquid.

2.9: Saturated Black Oil (Low Shrinkage Oil) Reservoirs

Saturated black oils are oils which are fully saturated with natural gas components (e.g., methane and ethane). This implies that the original reservoir temperature and pressure are in the two-phase region of the phase diagram at the time that the reservoir is discovered. This is shown in Figure 2.13.

Figure 2.13 (B) shows a saturated black oil in an anticlinal trap. In this figure, we can see that a gas cap overlies the oil at discovery. Since this gas accumulation was present at the original temperature and pressure conditions, we refer to this gas accumulation as a “primary gas cap.” Figure 2.13 (C) shows the state of the reservoir at abandonment conditions. As shown in this figure, at the lower pressure condition, the gas cap size has expanded.

2.10: Undersaturated Volatile Oil (High Shrinkage Oil) Reservoirs

Undersaturated volatile oil reservoirs are single-phase oil systems that reside in reservoirs with an original temperature relatively close to the critical point (relative compared to low shrinkage oils). As with the undersaturated black oils, the term undersaturated implies that the original reservoir temperature and pressure lies in the single-phase liquid region of the phase diagram. Such a system is shown in Figure 2.14.

We are now in a position to discuss the terms “low shrinkage oil” and “high shrinkage oil” as they pertain to the oil and gas industry. The blue bars on Figure 2.14 (A) represent equal pressure drops below the bubble-point pressure (these bars are of equal length). We can see from this figure that due to the convergent nature of the quality lines (dashed lines) near the critical point, equal pressure drops result in different volume percentages of the liquid phase in the two-phase region. The pressure drop represented by the blue bar further from the critical point results in a two-phase system with 90 percent of the volume occupied by the liquid phase; while the blue bar closer to the critical point results in a two-phase system with 80 percent of the volume occupied by the liquid phase. Thus, the volume of the liquid phase shrinks more in a system that is closer to the critical point for similar pressure drops. Since the liquid phase occupies less volume for a volatile, high shrinkage oil for comparable pressure drops, at abandonment the secondary gas cap will occupy a larger volume.

The definition of a “volatile fluid” is a fluid that easily evaporates or vaporizes. This definition gives rise to the name “volatile oils.” These systems develop larger gas phase volumes for comparable pressure drops when entering the two-phase region of the phase envelope.

2.11: Gas Condensate Reservoirs

Gas condensate reservoirs are gas systems that reside in reservoirs with the original temperatures lying between the critical temperature, T_C , and the cricondentherm (T_{max} in Figure 2.11). If the original reservoir pressure is greater than the dew-point pressure, then a single-phase gas system will occur in the reservoir as in Figure 2.15.

In this figure, the reservoir is initially a single-phase gas, but as we deplete the pressure due to gas production the pressure-temperature path of the system enters the two-phase region of the

phase envelope and liquid hydrocarbons condense from the gas in the reservoir (thus the name gas condensate reservoir.)

We are actually very familiar with this process in our everyday lives. Think of a four-component mixture of oxygen, carbon-dioxide, nitrogen, and water vapor (a mixture of air) and its phase envelope. If we started at a point in the single-phase gas region with a pressure below the cricondenbar (p_{max} in Figure 2.11) and reduced the temperature under constant pressure conditions (isobaric conditions), then we would cross the dew point locus of the phase envelope. This would be the pressure and temperature where we would see the first dew forming on plants and the first water condensation forming on glass and metal. If we were to continue reducing the temperature, then the percent volume of liquid (water) would increase and it would begin to rain. What is happening in this simple example is that the heaviest component in our system (water vapor) is condensing out of the gas phase (air mixture) and forming a second phase (liquid water) in the two-phase region.

This is essentially what is occurring in the reservoir of a gas condensate system but under isothermal conditions. As we pass through the dew point pressure, the heaviest hydrocarbon components in the system begin to drop out and form a second, liquid hydrocarbon phase in the two-phase region of the phase envelope inside the reservoir.

There is one interesting characteristic of gas condensate reservoirs that is worth further discussing and that is the retrograde behavior of these systems. This is illustrated in Figure 2.16. In this figure, if we were to follow the isothermal Path A-A'-A'', then we would go below through the dew-point pressure, increase the volume percentage of the liquid hydrocarbon phase until it reached a maximum at Point A' with further reductions in pressure resulting in a lower volume percentage of the liquid hydrocarbon phase. We could also continue the isothermal pressure reduction, reenter the single-phase gas region, and stop at Point A''.

We can see from Figure 2.09 and Figure 2.10 that for a pure (single-component) system, the liquid phase occurs at higher pressures than the gas phase. Thus, if we were to start in the single-phase liquid region of a pure system, we would need to reduce the pressure isothermally to create a gas phase.

The analogy for our multi-component system is that if we start at the point of maximum liquid volume (Point A' in Figure 2.16) and reduced the pressure isothermally, then we would get the conventional behavior for a pure system along Path A'-A''. Conversely, if we were to start at Point A' and increased the pressure isothermally to Point A in the single-phase gas region, then we would get the behavior opposite of that for a pure system along Path A'-A. This behavior, opposite to a pure system, is referred to as retrograde behavior. This behavior occurs in the green shaded region in Figure 2.16. This region, formed by connecting all of the points of maximum temperature on the quality lines, is referred to as the retrograde region of the fluid.

Sometimes we see or hear the terms “Retrograde Condensate Reservoir” or “Retrograde Condensate System.” This terminology is used to describe reservoirs in which the isothermal reservoir pressure-temperature path traverses the retrograde region of the system’s phase envelope.

2.12: Wet Gas Reservoirs

Wet gas reservoirs are gaseous hydrocarbon systems that reside in reservoirs with the original temperatures above the cricondenterm. In addition, the field surface facilities, separators, gas plant, etc., are in the two-phase region of the system's phase diagram. This is shown in Figure 2.17. Under these conditions, as the pressure in the reservoir is reduced due to production, the reservoir never enters the two-phase region of the gas and no hydrocarbon liquids drop out in the reservoir. That is, as pressure is reduced, the hydrocarbon fluid in the reservoir remains in its gaseous state. However, as the produced gas travels up the well and to the surface, at some point in the production system it enters the two-phase region of the phase envelope and liquid hydrocarbons develop in the well or surface facilities.

2.13: Dry Gas Reservoirs

Dry gas reservoirs are gaseous hydrocarbon systems that reside in reservoirs with the original temperatures above the cricondenterm. In addition, the surface conditions are also greater than the cricondenterm. This is shown in Figure 2.18. Under these conditions, as the pressure in the reservoir is reduced due to production, both the fluids in the reservoir and the fluids in the production system remain a single gaseous phase.

2.14: Key Learnings

- Crude oil and natural gas exist in the ground in the pore spaces of reservoir rock, typically sedimentary rock.
- Crude oil and natural gas accumulate in the ground in structures known as hydrocarbon reservoirs.
- There are five elements that are required for a hydrocarbon reservoir. The absence of just one element will preclude a geological feature from acting as a hydrocarbon reservoir. The timing of the formation of these elements is critical; all elements must be in place before the migration of oil or gas begins. These elements are:
 - o the source rock
 - o the migration path
 - o the cap rock
 - o the trap
 - o reservoir quality rock
- The source rock is an organic-rich sedimentary rock where catagenesis converts dead organic material, predominantly dead algae and zooplankton, into kerogen and eventually hydrocarbons.
- The migration path is a pathway where mature oil and gas migrate from the source rock to the trap. During the migration process, buoyancy is the dominant force acting on the hydrocarbons.
- The cap rock is an impermeable rock that prevents lighter oil and gas from migrating vertically away from the trap and towards the surface.
- The trap is a specific geological structure that keeps the crude oil and natural gas in place over geological time (i.e., provides four-way closure to the oil and gas). Traps can either be stratigraphic in nature, where the trapping mechanisms are dominated by the layering of

sedimentary rocks, or structural in nature, where the trapping mechanisms are dominated by the folding and faulting of the rocks.

□ The reservoir rock is a rock that has sufficient reservoir quality to allow for commercial crude oil and natural gas accumulations:

- o sufficiently high porosity to allow for the storage of crude oil and natural gas in commercial volumes;

- o Sufficiently high permeability to allow for the subsurface transport of crude oil and natural gas at commercial production rates.

□ The prevailing theory is that the hydrocarbons that make up crude oil and natural gas are organic in nature.

□ Crude oil and natural gas are complex mixtures of four hydrocarbon types:

- o alkane hydrocarbons

- o naphthenic hydrocarbons

- o aromatic hydrocarbons

- o Asphaltene hydrocarbons

□ The phase behavior of the complex mixtures of hydrocarbons are determined by the composition of the mixture and are governed by Gibbs Phase Rule.

□ The original pressure and temperature of a hydrocarbon reservoir at discovery are determined by the local hydrostatic gradient and thermal gradient, respectively.

□ Where the original pressure and temperature of the reservoir lies in the phase envelope of the phase diagram determines the type of crude oil or natural gas system that will be produced from the reservoir.

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Chapter 2

THE ORIGIN OF PETROLEUM

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1. HISTORIC OVERVIEW

Man has used petroleum since Biblical times, yet the origin of this natural resource remained a mystery for much of man's history. Classical literature is noticeably devoid of insight and even Roger Bacon laments in his 1268 treatise *Opus Tertium* on the lack of discussion on the origins of oils and bitumen by Aristotle and other natural philosophers.¹ Two theories on petroleum emerged during the Renaissance. In his 1546 text *De Natura eorum quae Effluunt ex Terra*, Georgius Agricola, a German physician, expanded on the Aristotelian concept of exhalations from deep within the Earth² and proposed that bitumen, like other minerals, condensed from sulfur. Andreas Libavius, another German physician, theorized in his 1597 text *Alchemia* that bitumen formed from the resins of ancient trees. These early discussions mark the beginnings of one of the longest running scientific debates: whether petroleum is formed by abiogenic processes that occur deep within the Earth, or from sedimentary organic matter that was once living organisms.

As fossil evidence emerged during the 18th century that coals were derived from plant remains, many scientists proposed similar origins to explain petroleum. The historic record is somewhat questionable, but Mikhailo Lomonosov is credited by some to have proposed the theory that liquid oil and solid bitumen originate from coal through underground heat and pressure as early as 1757³ and certainly by 1763⁴. Various biogenic theories emerged during the early 19th century suggesting that petroleum was derived directly from biological remains or through a distillation process⁵.

Modern theories that petroleum originated from ancient sedimentary, organic-rich rocks emerged during the 19th century. T.S. Hunt of the Canadian Geological Survey concluded in 1863 that the organic matter in

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some North American Paleozoic rocks must be derived from marine vegetation or marine animals, and that the transformation of this organic matter to bitumen must be similar to the processes involved in coal formation⁶. Leo Lesquereux, the American father of paleobotany, reached similar conclusions after studying Devonian shales in Pennsylvania⁷, as did Newberry in his study of Devonian shales in Ohio.⁸ Early 20th century field⁹ and chemical¹⁰ studies of the Monterey Formation by the U.S. Geologic Survey provided convincing evidence that the oil was derived from diatoms in the organic-rich shales. Similar studies of organic-rich shales conducted in Europe during this time arrived at the same conclusion.^{11,12}

Full ascendancy of the biogenic hypothesis began in the mid-20th century with a convergence of scientific advances in paleontology, geology, and chemistry. In 1936, Alfred Treibs established a link between chlorophyll in living organisms and porphyrins in petroleum.¹³ Additional geochemical evidence followed with the discoveries that low to moderate maturity oils still retained hydrocarbon fractions with optical activity¹⁴, that the stable isotopes of carbon of petroleum bear a biological fractionation¹⁵, and that oils contain in addition to porphyrins, a host of hydrocarbons that can be traced back to specific biological precursors¹⁶. Concurrent with these findings were field studies recognizing that organic-rich strata occur in all petroliferous sedimentary basins, that this sedimentary organic matter (kerogen) is derived from biota, that it has been chemically altered from its initial state^{17,18} and, that oil and gas is produced from this kerogen as the sediments are buried and heated.¹⁹

Although there is overwhelming evidence for a biogenic origin, some still advocate abiotic theories. The development of the modern abiogenic concept is rooted in the mid-19th century. The prolific French chemist Marcelin-Pierre Berthelot described in 1860 experiments where n-alkanes formed during the acid dissolution of steels.²⁰ Dmitri Mendeleev reasoned in 1877 that surface waters could percolate deep within the Earth, react with metallic carbides forming acetylene, which could then condense further into larger hydrocarbons.²¹ Mendeleev's abiotic theory, further refined in 1902²², was viewed initially as particularly attractive as it offered an explanation for the growing awareness of the widespread occurrence of petroleum deposits that suggested some sort of deep, global process.

Advocates of abiotic origin theories dwindled under the mounting evidence for a biogenic origin of petroleum. By the 1960's, there was little support for an abiotic origin, except among a small group within in the Former Soviet Union. First proposed in 1951 by Nikolai Kudryavtsev²³ and advanced over the years in numerous Soviet publications,³ a modernized version of Mendeleev's hypothesis emerged. This theory relies on a thermodynamic argument, which states that hydrocarbons greater than methane cannot form spontaneously except at the high temperatures and pressures of the lowest most crustal depths. The theory ignores the fact all of life relies on being in thermodynamic disequilibrium with its environment. In the West, astronomers have been the most vocal advocates for abiotic petroleum. Carbonaceous chondrites and other planetary bodies, such as asteroids, comets, and the moons and atmospheres of the Jovian planets, certainly contain hydrocarbons and other organic compounds that were generated by abiotic processes.²⁴ Sir Fredrick Hoyle reasoned in 1955 that as

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the Earth was formed from similar materials, there should be vast amounts of abiogenic oil.²⁵ In more recent years, Thomas Gold is the strongest promoter for abiotic petroleum.^{26,27} Against the advice of nearly all geochemists and petroleum geologists, Gold convinced the Swedish government to drill two deep wells (Gravberg 1 in 1986-1990 and Stenberg 1 in 1991-1992) into fractured granite under the Siljan ring, the site of an ancient meteorite crater. The wells failed to find economic reserves and evidence for even trace amounts of abiotic hydrocarbons is controversial.²⁸

Geochemists do not deny the existence of abiogenic hydrocarbons on Earth. Small amounts of abiotic hydrocarbon gases are known to be generated by rock-water interactions involving serpentinization of ultramafic rocks,^{29, 30} the thermal decomposition of siderite in the presence of water,³¹ and during magma cooling as a result of Fischer-Tropsch type reactions.³² However, commercial quantities of abiotic petroleum have never been found and the contribution of abiogenic hydrocarbons to the global crustal carbon budget is inconsequential.³³

2. THE PETROLEUM SYSTEM

The accumulation of economic volumes of petroleum (oil and/or gas) in the subsurface requires that several essential geological elements and processes be present at specific time and space.^{34,35} Source rocks generate and expel petroleum when sufficient thermal energy is imparted to the sedimentary organic matter (kerogen) to break chemical bonds. This heating is induced usually by burial by overburden rock. Once expelled, petroleum migrates either along faults and/or highly permeable strata. Accumulations form only when high porosity strata (reservoir rocks) are charged with migrating petroleum and the petroleum is prevented from further migration. These petroleum traps are formed only when geologic movements result in subsurface topographies (structural and stratigraphic) that block migration and when the reservoir rocks are covered by low permeability strata (seal rocks). The mere presence of these geologic elements is insufficient to form petroleum reserves. Traps must be available at the time of oil expulsion and, once charged, their integrity must be preserved until exploited. These elements and processes constitute the Petroleum System (Figure 1).

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overburden

anticlinal seal

trap reservoir oil migration

source gas migration

underburden

+ + basement stratigraphic

trap

Fault-bound

traps

Top Oil Window

Top Gas Window

Figure 1. Elements of a Petroleum System. All petroleum systems contain: 1. at least one formation of organic-rich sediments that has been buried to a sufficient depth by overburden rock such that petroleum is generated and expelled, 2. Pathways (permeable strata and faults) that allow the petroleum to migrate, 3. Reservoir rocks with sufficient porosity and permeability to accumulate economically significant quantities of petroleum, and 4. Sealing rock (low permeability) and structures that retain migrated petroleum within the reservoir rock. The top and bottom of the oil window is approximated as a function of burial depth. In actual basins, these depths are not uniform and vary as a function of organic matter type, regional heat flow from basement, in thermal conductivity of the different lithologies, and burial history (e.g., deposition rates, uplift, erosion, and hiatus events).

A rigorous discussion of the origin of petroleum should encompass all of the interrelated elements of the petroleum system. Such a discourse is beyond the scope of this review, which will focus only on the deposition of organic-rich strata and the generation of petroleum from these sources. The reader is referred to *Exploring for Oil and Gas Traps*, a publication of the American Association of Petroleum Geologists, for a complete discussion of all aspects involved with defining the petroleum system for effective exploration.³⁶ Books by Tissot and Welte³⁷ and Hunt³⁸ provide detailed discussions of the principles of petroleum geochemistry, and a recent paper by Peters and Fowler³⁹ is an excellent review of the application of modern geochemical techniques to exploration and production practices.

3. DEPOSITION OF ORGANIC-RICH SEDIMENTARY ROCKS

Petroleum source rocks are water-deposited sedimentary rocks that contain sufficient amounts of organic matter to generate and expel commercial quantities of oil and/or gas when heated. Such organic-rich strata were deposited throughout Earth's history, in nearly all geologic environments, and in most sedimentary basins. Source rocks, however,

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typically represent only a minor amount of basinal strata and are formed only when specific conditions exist.

Three general factors control the deposition of organic-rich sediments: productivity, dilution, and preservation (Figure 2).^{40,41,42} Biological productivity determines the amount of organic matter that is contributed to sediments. Dilution refers to the amount of inorganic minerals that mixes with the organic matter. Once deposited, the organic matter must be

preserved in a form that may later generated petroleum. There was once an active debate as to which factor was the most important in forming organic-rich sediments.⁴³ It is now recognized that these three factors are inherently interrelated in a highly complex, and variable manner.

Biological

Productivity

photosynthesis

or x

sp Flu

Se iotic

B

t

co In

an nt

Tr trie

nd pu

Sedimentary

ar t

u

N

y

Organic

Carbon

Sediment Preservation

Dilution

Aerobic/Anaerobic

Terrigenous clastics Decomposition

Uplift

evaporates Redox conditions

Erosion

Skeletal matter

Climate, Sea Level,

Tectonics, Volcanism,

Hydrates

Figure 2. Three major factors, primary productivity, preservation, and dilution, determine whether organic-rich source rocks are deposited. These factors are interrelated and influenced by a number of geologic conditions.

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Photosynthetic organisms, which include aerobic cyanobacteria, algae,

phytoplankton, land-plants, and some anaerobic bacteria, provide most of the initial organic matter by fixing CO₂ into biomass. The contribution of organic matter by non-photosynthetic chemotrophs is minor, except in some unusual environments, such as the deep-sea hydrothermal vents. Most of the non-photosynthetic biosystems, such as methanotropic communities⁴⁴, rely on recycled carbon that was fixed originally by photosynthetic organisms. Many factors moderate the biota and the primary productivity (e.g., such as nutrient input from rivers and coastal upwellings, pCO₂, temperature, and turbidity) that are influenced by global climatic and tectonic conditions.

Figure 3. Generalized redox cycle for organic carbon. Production of new organic matter by the photosynthetic fixation of CO₂ (primary productivity) can occur with (aerobic) or without oxygen (anaerobic) as a byproduct. Respiration and other processes result in the nearly complete oxidation of this organic matter back to CO₂. A small amount of the organic matter in sediments escapes biological recycling and is preserved in rocks. Eventually, this carbon is recycled to the surface as CO₂ by geologic processes, such as subduction and venting, erosion and weathering – or more recently, by the combustion of fossil fuels.

Much of the primary organic matter created by photosynthetic or chemosynthetic autotrophs undergoes degradation by other organisms during the secondary production of organic matter. Heterotrophic organisms in the water column, sediments, and rock continually degrade and rework primary aquatic and terrigenous organic matter (Figure 3). Aerobic respiration is very rapid and efficient and primary organic matter may pass through a chain of

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water and mud dwelling animals, protozoa, and bacteria, until fully consumed and returned to the atmosphere/hydrosphere as CO₂. Anaerobic bacteria also degrade organic matter either by fermentation or by respiration using terminal electron acceptors other than O₂ (e.g., nitrate, sulfate, iron). These microbial metabolisms are generally slower and may be curtailed by limited nutrient and electron acceptors within the sediment porewaters. Consequently, some of sedimentary organic matter, both from primary and secondary biogenic sources, may escape recycling and become preserved in lithified rock. Hence, anoxic conditions enhance the preservation of oil-prone organic matter and promote the deposition of potential source rocks (Figure 4).

Figure 4. Oxidic (left) and anoxic (right) depositional environments generally result in poor and good preservation of deposited organic matter, respectively (after Demaison and Moore, 1980⁴⁵). The solid horizontal line separates oxidic (above) from anoxic (below). In oxidic settings, bottom dwelling metazoa bioturbate the sediments and oxidize most organic matter. In anoxic settings, especially where the oxidic-anoxic boundary occurs in the water column, bottom-dwelling metazoa are absent and sediments are not bioturbated.

The amount of oxygen in bottom waters or sediments is determined by it

rate of influx from the photic zone (via circulation and/or diffusion) and its rate of consumption (biological oxygen demand). Topographical barriers, currents, or water stratification due to temperature and salinity gradients may limit water circulation and diffusion. A high supply of primary organic matter raises the biological oxygen demand, rendering pore waters, and even the lower water column, anoxic. This mechanism is seen in terrigenous environments with high input of land-plant material (e.g. swamps, and coastal
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plains), and in lake and marine environments with high nutrient influx (e.g, seasonal rains and overturn of water columns, and upwelling zones). In anoxic marine environments, the most prevalent organisms are bacteria that utilize sulfate and produce H_2S . If not precipitated as pyrite by iron from clays and other clastic minerals, H_2S may react with organic matter, incorporating sulfur. If the production of H_2S is greater than its rate of sequestering, euxinic conditions (waters with free H_2S) occur. The transition between oxic and euxinic waters may occur within the sediments or water column. Some strata with excellent source potential were deposited where euxinic conditions extended into the photic zone.⁴⁶ Biological activity may also be restricted by hypersaline conditions that occur in playa lakes, lagoons, and restricted marine settings.

Sedimentation rate influences both the preservation and concentration of organic matter. The inorganic materials may be clastic (eroded clays and sands), chemical precipitates (carbonates, salts), or biogenic (siliceous and carbonate shells). Organic matter may be removed from biological recycling by rapid burial, such as in deltaic settings. However, preservation is offset by dilution and the resulting rocks may be relatively low in organic carbon. Similarly, sediments resulting from high primary productivity, such as diatomaceous cherts, may have an upper limit in organic carbon because of high autodilution. In general, sedimentary rocks with the highest concentrations of organic matter are deposited under conditions of moderate to high influx of primary organic matter, anoxic (possibly euxinic) bottom waters, and low sedimentation rates.

4. KEROGEN FORMATION AND THE GENERATIVE POTENTIAL OF SOURCE ROCKS

Although derived from biochemicals, the sedimentary organic matter in source rocks, is markedly different in structure and chemical composition. Living organisms are composed mostly of proteins, nucleic acids, and lipids. Structural materials, such as lignin and cellulose, and various resins are fairly well-defined biopolymers composed of a limited number of monomers. With the exception of some halophilic green algae and a few bacteria⁴⁷, only trace amounts of hydrocarbons are produced directly as biomass by most living

organisms. In contrast, immature sedimentary rocks contain only low concentrations of the functionalized biochemicals. Most of the organic carbon is bound in a condensed, insoluble macromolecular material, termed kerogen, which has comparatively few functional groups. Kerogen has been called a “geopolymer” but this is a misnomer, as the term implies that there is a repetition of distinct monomers and some structural order. Kerogen has no unique molecular structure and can only be defined in terms of bulk elemental composition and average molecular distributions.

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BIO SYNTHESIS

Microbial Recycling Microbial Recycling

BIOMASS

Proteins, Low-Molecular

Polysaccharides Weight Biomacro- Functionalized

Hydrocarbons

Nucleic acids Biomolecules molecules lipids

CO₂, NH₃ Selective Selective Selective Selective

Preservation Preservation Preservation Preservation

Hydrolysis

Microbial

DIAGENESIS

Degradation

Bioresistive

macro-

molecules

Sulfurization Sulfurization

Amino acids Resistive

sugars lipids

nitrogen bases

Condensation

Defunctionalization Condensation

Bitumen Bitumen

Humic acids

Fulvic acids

Humin

CATAGENESIS

Kerogen

Thermal Thermal

Decomposition Decomposition

Thermal

Decomposition

Hydrocarbons

NSO compounds

Figure 5. Kerogen is formed during diagenesis by the selective preservation of bio-resistive macromolecules and from the incorporation of lower-molecular weight species. Modified from Tegelaar et al. (1989)⁴⁹.

Optically, kerogen is often described as mixtures of amorphous organic matter and macerals, morphologically distinct particles that are mostly derived from land-plants. The amorphous matter was once thought to be derived from the complete breakdown of biological macromolecules that then re-assembled in a random fashion with low molecular-weight biochemicals (e.g., lipids).⁴⁸ We now realize that there are biological macromolecules that are alkyl-rich and resist microbial recycling.⁴⁹ These bio-macromolecules (e.g., cutan, sporopollenin, tannins, and algaenan) may constitute only a small fraction of the initial biomass, but are selectively preserved and enriched during early diagenesis (Figure 5). They provide a core for the incorporation of functionalized low molecular-weight biochemicals, such as membrane lipids and the breakdown products of less resistant bio-macromolecules. A small fraction of this material may remain soluble, which along with biogenic

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hydrocarbons and lipids that resist incorporation, becomes the bitumen that can be extracted from immature source rocks using organic solvents.

Kerogen formation begins at the point when living cells die and their biochemicals are exposed to the geologic environment. Microbial processes are mostly responsible for the breakdown of biological macromolecules and the recycling of lower molecular-weight biochemicals, resulting in the selective preservation of the more bio-resistive compounds. Low temperature chemical reactions further alter kerogen through the loss of functional groups (e.g., $-NH_3$, $-COOH$), sulfur incorporation, condensation, cross-linking, and aromatization. The processes and thermal regime under which kerogen forms, termed diagenesis, occurs under mild thermal conditions ($<80^\circ C$).

1000

Rock-Eval Hydrogen Index

World Class

Oil Prone

Source Rocks

Range of Typical

500

Oil-prone

Source Rocks Oil Shales

1.2

Atomic H/C

Gas & 200

Oil Prone

0.8 100

Gas Prone

Coaly

Shales

Poor Moderate Good Excellent

Potential Potential Potential Potential

0.5 1 2 5 10 20 50

%TOC

Figure 6. The quantity and quality of kerogen determines the generative potential of a source rock. Boundaries for the classifications of generative potential are approximate. Typical marine source rocks from productive basins contain ~2 to 6% TOC. Basins with world class reservoirs may contain source units with appreciably higher amounts of carbon. Expulsion of oil may be difficult from rocks with <1-2% TOC. The hydrogen content of the organic carbon determines the quality of the expelled hydrocarbons (during metagenesis). Kerogens with high H/C ratios tends to generate oil, while those with low H/C ratios tend to generate gas. The Hydrogen Index is based on the Rock-Eval instrumentation. This pyrolysis method, commonly used by petroleum geochemists to rapidly screen rocks for their generative potential, heats a ground sample and measures the pyrolyzate response on a flame ionization detector and a CO₂ response by an infrared detector. These responses can be calibrated to a rough approximation of the kerogen H/C and O/C atomic ratios.

The quantity and quality of organic matter preserved during diagenesis of sediment determines the generative potential of the source rock and whether it will be prone to expel oil or gas. Quantity is determined by amount of organic input, the degree to which it is preserved (either as primary or

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secondary biogenic matter), and by its dilution with inorganic mineral matter. Typical oil-prone, marine source rocks contain 2-5 wt.% total organic carbon (TOC), but strata in the range of 10-20% are known to occur in high yield petroleum systems (Figure 6). Oil shales, many of which are lake deposits, contain over 10% TOC. Coals are >80% TOC, and may be dilute with varying amounts of clastic minerals to yield coaly shales with a wide range of carbon content.

Kerogen quality is mainly a function of hydrogen content – kerogens with high H/C ratios (> 1.2) are oil-prone, while those with lower H/C ratios (0.5 to 0.8) tend to generate mostly gas. Biogenic input of organic matter and the

manner of its preservation determine the hydrogen content of kerogen. Some algae and bacteria are extremely rich in membrane lipids and aliphatic biopolymers, such as algaenan. Sediments that receive this input have the potential to be hydrogen-rich (kerogen H/C > 1.5). Conversely, land-plant lignins have low H/C ratios and sediments that received only this input are hydrogen-poor (kerogen H/C ratios < 1.0). The hydrogen, sulfur, and oxygen content of kerogen can be modified greatly during deposition and diagenesis. Oxidic conditions favor rapid and fairly complete heterotrophic consumption of primary organic matter. The organic matter remaining in the sediments tends to be oxidized or inert. Anoxic conditions conserve, or even enhance, the initial H/C ratio of the primary organic matter via selective preservation and/or contributions from secondary biota. Euxinic conditions promote the incorporation of sulfur, resulting in high S/C ratios.

There are many schemes that attempt to classify kerogen types by their morphology, maceral and palynological assemblages, and/or bulk chemical composition. The most widely used is a modification of a method developed for coals⁵⁰, whereby the H/C and O/C ratios are plotted (Figure 7). Kerogens are assigned designations as being Type I, II, III, or IV depending where they fall on the plot. Type I and II are oil-prone, Type III is gas-prone, and Type IV is inert carbon.

This classification scheme can be sub-divided further by considering the variation in sulfur content (Table 1). Since the incorporation of sulfur into sedimentary organic matter involves the generated of H₂S by sulfate reducing bacteria, kerogens with high S/C ratios (0.04 to > 0.10) are found mostly in marine (non-clastic) rocks where euxinic conditions may prevail. Type I kerogens are deposited typically in continental lakes, which have no contact with marine waters and lack a ready supply of sulfate. Consequently, sulfur-rich Type IS kerogens arise only in unusual geologic settings where older evaporite beds (e.g., gypsum or anhydrite) are exposed and dissolved into normally fresh lake water.⁵¹

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Figure 7. van Krevelen type diagram showing the distribution of kerogen types and some of their precursor macerals in relationship to their H/C and O/C atomic ratios. A substantial portion of their oxygen content is lost during diagenesis. Oil generation occurs during catagenesis, whereby the H/C ratio of the residual kerogens decrease. When the H/C ratio is < 0.5, the residual kerogens are capable of generating only methane.

Type II kerogens are deposited primarily in marine settings dominated by photosynthetic organisms. During early diagenesis, this organic matter may react with H₂S produced by sulfate-reducing bacteria that thrive in marine, anoxic waters and sediments. H₂S also reacts with iron and other metals, forming inorganic sulfides. Consequently, marine Type II kerogens may

have varying sulfur content. Marine shales and mudstones, which contain iron oxides and iron-rich clays, yield low-sulfur Type II kerogens and pyrite. Carbonate, evaporite, and chert deposits, which are iron-poor, yield high-sulfur Type IIS kerogens.

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Type III kerogens form in swamps and coastal plains. As these depositional environments may or may not have marine influence, the sulfur content of coals can vary, but is generally low. Type III kerogens are frequently allochthonous; that is, they are transported by fluvial systems into delta and nearshore marine sediments where they may mix with locally produced (autochthonous) Type II kerogens. Most Type III kerogens are dominated by vitrinite, a land-plant maceral that has a low H/C ratio and is gas-prone. Some coals, however, contain oil-prone material (Type IIIC) that are derived from associated algae or from the selective preservation of cutinous macromolecules. Many oils from Southeast Asia are generated from Tertiary coaly shales that contain this type of kerogen.

Table 1. Kerogen types, their occurrence and bulk chemistry (at end of diagenesis)

Type Depositional Setting Primary Biotic H/C O/C S/C

Input

I Lakes, restricted green algae $> 1.4 < 0.1 < 0.02$

lagoons cyanobacteria

dinoflagellates

IS Lakes with a source of green algae $> 1.4 < 0.1 > 0.04$

sulfate (rare) cyanobacteria

II Marine shales marine algae, $1.2 - 1.4 \sim 0.1 \ 0.02 - 0.04$

phytoplankton

IIS Marine carbonates, marine algae, $1.2 - 1.4 \sim 0.1 > 0.04$

evaporites, silicates phytoplankton

III Coals, coaly shales, vascular land- $0.7 - 1.0 > 0.1 < 0.021$

deltas plants

IIIC Coastal plains (oil- vascular land- $1.0 \text{ to } 1.2 > 0.1 < 0.021$

prone coals) plants, algae

IV Inert carbon due to All possible $< 0.5 < 0.15 < 0.022$

oxidation or advanced

maturity

1

Coals generally are low in sulfur. Depositional settings with marine influence may result in coals with $S/C > 0.02$.

2

Some inert pyrobitumens may have higher S/C ratios as a result of secondary sulfur incorporation.

5. GENERATION AND EXPULSION OF OIL AND GAS

During catagenesis, kerogen thermally cracks and produces bitumen. Weak C—S and C—O bonds break preferentially during the early stages of catagenesis producing bitumen that is highly enriched in polar (NSO) compounds. With additional heating (~90-140°C), C—C bonds break within the evolved polar compounds and from residual kerogen, yielding a hydrocarbon-rich fluid that is then expelled from the source rock's mineral matrix. With additional thermal stress, kerogen yields primarily condensate and then wet gas (C1-C6). Catagenesis is complete when the kerogen has expended its capacity to generate C2+ hydrocarbons (~150-175°C).

Metagenesis may then take place under still more severe thermal alteration,

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where only methane is produced as methyl-groups are cleavage from highly condensed, aromatic structures.

H/C

100 1.2

25

20 Type II

1.1

80 15

10

1.0

Kerogen 5

% Fraction

60 0.9 0

42 44 46 48 50 52 54 56 58 60 62 64 66

0.8

40

1°C/Ma heating rate Heavy Oil

Methane

0.7

Light Oil Wet Gas 4.0

20

0.6

1.3 1.8 3.6

0 0.5

30 80 130 180 230 280

Temperature, °C

Figure 8. Example of a kerogen decomposition kinetic model with predictions of expelled fluid

composition. The rate of kerogen conversion is assumed to follow a series of parallel first order kinetic reactions defined by the Arrhenius equation, $k = Ae^{-(E/RT)}$, where k is the rate coefficient, A is a constant termed the frequency factor, E is the activation energy, R is the universal gas constant, and T is the temperature. The kerogen model was derived by fitting the yield curves of pyrolyzates that evolved from a marine shale source rock containing Type II kerogen heated at several different heating rates under anhydrous conditions. In this case, the model was fitted using a fixed frequency factor, $A = 1 \times 10^{14} \text{ sec}^{-1}$. The evolved products are predicted from a more advanced model that accounts for expulsion and secondary cracking of retained bitumen and residual kerogen.

Oil generation is a kinetic process – both time and temperature are critical⁵². The kinetic parameters of kerogen decomposition can be determined by artificially heating immature source rocks using relatively fast heating rates (~ 300 to 550°C at ~ 1 to 50°C/min) or lower isothermal temperatures (275 – 350°C for several days) and measuring either product yield or the residual generative potential. Data are fitted to kinetic models defined by a series of first order parallel reactions, usually at a single frequency factor. When geologic heating rates are applied (~ 1 to 10°C/Ma), the derived kinetic models yield kerogen maturation results comparable to what is observed in

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petroliferous basins.^{53,54} More advanced models describe the fluid composition of the expelled products and include secondary cracking reactions (Figure 8).

Although simple thermogenesis adequately describes petroleum generation, geocatalysis involving reactive mineral surfaces, clays, trace metals, or organic species have been proposed.^{55,56} Such processes are highly speculative, and it is difficult to image how inorganic agents would remain activated under subsurface conditions or how mass transport limitations inherent in solid-solid interactions could be circumvented⁵⁷.

Once generated from the kerogen, petroleum is expelled into mineral pore spaces (primary migration) and then through permeable rock and faults to the trap (secondary migration). Oil expelled from a source rock is enriched in saturated and aromatic hydrocarbons relative to the bitumen that remains (Figure 9). The factors that control primary expulsion and bitumen fractionation are largely unknown; and, although there is generally a lack of experimental or observational evidence for the underlying mechanisms, there is no shortage of hypotheses. Expulsion models that attempt to account for these chemical differences generally assume rate-limiting processes occur in the release of generated hydrocarbons from the kerogen or in the movement of hydrocarbons within the mineral matrix.

Hypotheses based on kerogen-oil interaction postulate that the expulsion of oil is controlled by absorption or adsorption of the products onto the

surface of the kerogen,^{58,59,60} diffusion of the hydrocarbons through the kerogen^{61,62,63,64} and/or relative solubility.⁶⁵ These hypotheses attribute little importance to movement of petroleum within the source rock mineral matrix and the efficiency of the release of oil is controlled primarily by the amount of organic carbon and its composition (see, Pepper & Corvi, 1995; and references therein⁵³). Expulsion models based on the interactions of generated products with the source kerogen have gained favor in recent years as they have the potential to account for compositional differences between bitumens and expelled oil and for expulsion efficiencies that depend on kerogen type and richness. This thinking can be traced to observations made on source rocks⁶⁶ and to recognition of the absorptive capacity of solid organic matter as revealed by solvent swelling experiments.^{67,68,69,70} It is now clear that Type I and II kerogens and Type III and IIIC coals have sufficient sorptive properties to explain residual oil concentrations in mature source rocks.

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Source Rock

Bitumen Extracts Expelled Oils

Type II

(Marine Shales)

Type IIS

(Carbonates)

saturates

aromatics + thiophenic-aromatics

polars (NSO) + asphaltenes

Figure 9. Comparison of C₁₅+ chemical group type distributions of expelled oil and retained bitumens for typical Type II (marine shale) and Type IIS (carbonate) source rocks (middle of the the oil window).

There are many expulsion models that target chemical or physical processes of oil moving within the source rock mineral matrix as the rate determining step. Different aspects have been considered to be important in this movement. Some consider the amount and type of organic matter as being critical to generating sufficient bitumen to exceed a pore saturation threshold.^{71,72,73} Others postulate that the establishment of effective and continuous migration pathways within the source rocks is critical.^{74,75,76} Other factors suggested to be key elements in expulsion are: pressure build-up from generation and compaction and the failure of the rock fabric resulting in micro-fracturing,⁷⁷ gas availability and movement of oil in a gas or supercritical phase,^{78,79} or movement of oil in an aqueous phase.^{80,81} For the most part, the conditions that determine these elements are controlled by

the primary sedimentological conditions in the depositional environment of the source rock and secondary diagenetic processes. Consequently, the

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mechanisms that define oil movement differ according to the lithofacies of the source rock.

6. COMPOSITION OF PRODUCED PETROLEUM

The molecular and isotopic composition of produced petroleum is determined by complex chemical, physical, and biological processes.

Generation and expulsion from the source rocks, phase behavior as the petroleum moves from source to reservoir, reservoir fill history, and secondary alteration processes all influence oil and gas compositions.

Each source facies generates oil with distinct chemical composition that reflects biotic input, depositional setting, and thermal history (Figure 10). For example, lacustrine and coaly source rocks generate waxy, low-sulfur crudes, while carbonate and evaporite source rocks generate asphaltic, high-sulfur crudes. These compositional differences are most apparent during the initial stages of oil expulsion and become less distinct as the source proceeds through catagenesis where secondary cracking reactions become prevalent. Although petroleum is derived from biological organic matter, most of the individual compounds cannot be assigned to a specific biochemical precursor. Some petroleum hydrocarbons, termed biomarkers, retain enough of their original carbon structure that a likely biochemical precursor can be assigned.⁸² The abundance and distribution of biomarkers allow geochemists to infer the origin and thermal history of oils.

Once generated and expelled from the source rock, petroleum composition can be further modified during migration and entrapment within the reservoir. In most petroleum systems, the source formation is at greater temperature and pressure than the reservoir and migrating petroleum fluid may separate into gas and liquid phases that can then migrate independently.^{83,84,85} Petroleum also interacts with water and the more soluble hydrocarbons may selectively partition into the aqueous phase.⁸⁶

Once in the reservoir, secondary processes can alter oil composition. Biodegradation, the consumption of hydrocarbons by microorganisms, is likely to occur in shallow, cool reservoirs (<80°C).^{87,88} This process selectively removes saturated hydrocarbons, enriching the residual crude oil in polar and asphaltic material. Biodegradation forms acids and biogenic CH₄, CO₂, and H₂S. Microbial alteration of crude oils is a relatively fast process and may occur naturally or result from poor production practices.

Thermochemical sulfate reduction (TSR) is another reservoir alteration process that can affect oil quality and quantity. It is a chemical redox process that occurs at relatively high temperatures (>120°C), where hydrocarbons are

oxidized to CO₂ and sulfate is reduced to H₂S.^{89,90} The residual oil is depleted in saturated hydrocarbons and enriched in sulfur-aromatic species. Reservoir charging and fill history also can alter oil composition. For example, mixing

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of gaseous hydrocarbons with a heavy oil can cause asphaltenes to precipitate, forming a tar mat.⁹¹

Type IIS carbonate source Type II marine shale source

API = 17.6 %S = 4.48

Ni = 56 ppm V = 320 ppm API = 39.4° %S = 0.18

n-C5

n-C8

n-C4

%Sat = 12.4 %Aro = 28.4 Ni = 3 ppm V = 5 ppm

n-C9

n-C17

Methycyclohexane

%NSO = 37.7 %Asph = 19.7 %Sat = 66.4 %Aro = 24.6

n-C6

18

n-C7

Phytane n-C

Pr/Ph = 0.62 CPI = 0.85 %NSO = 8.3 %Asph = 0.9

n-C7

n-C10

n-C17

Pr/Ph = 1.55 CPI = 1.01

n-C8

n-C18

Pristane

n-C9

n-C11

Pristane

n-C10

n-C11

n-C12

n-C12

Phytane

n-C13

n-C13

n-C15

n-C14

n-C 14

n-CC1716

n-C15

n-C19

n-

n-C16

18

n-C6

n-C20

n-C27n-C28

n-C

n-C21

n-C17

n-C18

n-C31 n-C30

n-C22

n-C26

Phytane

n-C23

24

n-C19

n-C

n-C20

26

n-C21

n-C29

Pristane

n-C

n-C22

32

Pristane

n-C23

n-C

Phytane

n-C24

n-C25

n-C34

n-C26

n-C33

ip16

ip13

ip18

ip14

ip15

ip18

Type I lacustrine source

Type IIIC coaly source

API = 41.3 %S = 0.07

API = 38.0 %S = 0.17

Ni = 4 ppm V = 3 ppm

NI = 8ppm V = 36 ppm

%Sat = 68.7 %Aro = 13.0

%Sat = 49.0 %Aro = 28.0

Pristane

%NSO = 17.8 %Asph = 0.6

%NSO = 20.3 %Asph = 0.7

Pr/Ph = 2.63 CPI = 1.05

Pristane

Pr/Ph = 6.23 CPI = 1.11

n -C15

n -C14

C1

n-

n -C 17

n -C 17

n -C18 6

n -C 18

n -C 13

n -C 12

n -C 19

n -C 11

n -C 21

n -C 20

n-C18

n -C10

n -C 22

n -C 23

n-C17

n -C 25

Phytane

n -C 24

n -C 9

Pristane

n-C25

n -C 26

n-C23

n-C24

n-C26

n-C16

n-C22

n -C 27

Phytane

n-C21

MCH

n -C8

n-C27

n-C19

n-C20

n-C18

n-C14

n-C15

n-C28

n-C29

n -C29 28

n-C13

n -C

n-C12

n-C31

Pristane

n-C30

n -C 7

n-C17

n-C11
 n -C31 30
 n-C10
 n -C
 n-C32
 n-C34 n-C33
 Phytane
 n-C9
 n -C 33 32
 n -C
 n -C 6
 n -C34
 n-C8
 n-C7
 n-C6

Cadinanes

Figure 10. Whole-oil gas chromatograms showing examples of oils from different source kerogens. The grayed area, showing the elution of n-C17, pristane (Pr), n-C18, and phytane (Ph), is enlarged. Pr/Ph is the pristane/phytane ratio, which is used to infer the depositional environment (anoxic, hypersaline Pr/Ph<1, oxic water column/anoxic sediments Pr/Ph 1~2, oxic deposition Pr/Ph>2). CPI is the carbon preference, which is used to infer carbonate/evaporite source facies <1, and higher land-plant input CPI > 1. API gravity, the concentrations of sulfur (S), nitrogen (N), nickel (Ni) and vanadium (V) and the C15+ chemical group-type compositions are influenced by of the organic matter, depositional environment, and thermal history of their source rocks and by reservoir alteration processes.

The composition of crude oil that arrives at a refinery is not identical to reservoir fluids. Gas and water is separated at the well head and emulsions are broken. Consequently, oils lose some light hydrocarbons (<C6) by evaporation during production and transport. Pipeline and tanker oils are frequently blends of oils from multiple fields and reservoirs, which individually may be of varying composition and quality. For these reasons, testing of subsurface fluids from individual reservoirs is necessary to determine field economics and design reservoir management practices.

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7. SUMMARY

The theory that petroleum originates from sedimentary organic matter that was once living organisms is consistent with all natural observations, laboratory analyses and experiments, theoretical considerations, and basin simulations. The accumulation of economic quantities of petroleum (oil and

gas) requires that a series of processes occur within sedimentary basins. Organic-rich sediments are deposited only under specific conditions that promote the production of biota and/or transport of biogenic organic compounds and the selective preservation of this material. The sedimentary organic matter converts to kerogen, an insoluble macromolecule with a composition that reflects the biotic input and chemical alterations (sulfurization, condensation, defunctionalization, and aromatization) that occur during diagenesis. Once lithofied, these organic-rich strata have the potential to generate oil and gas when buried and heated to promote thermal cracking. Expelled petroleum migrates from the source through fractures and permeable strata. Economic reserves occur when geological conditions allow for the accumulation, retention, and preservation of significant volumes of migrated petroleum. Collectively, these processes describe a petroleum system.

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